

# A Unified Nonparametric Framework for Cognitive Diagnosis with Polytomous Data and AI-Assisted Q-Matrix Construction

## PROJECT DESCRIPTION

### Overview

The proposed research project concerns the development of a **unified framework for nonparametric cognitive diagnosis (NPCD) for polytomous data**. NPCD refers to an approach to CD that avoids parametric statistical models requiring computationally expensive estimation. Instead, examinees' skill profiles are identified through discrete combinatorial algorithms using a distance-based loss function, which eliminates reliance on large calibration samples. These nonparametric classification algorithms are computationally feasible even with very small samples. In this proposal, polytomous data refers to CD assessment settings where either (i) **item responses are dichotomous while the underlying attributes are polytomous**, or (ii) **item responses are polytomous while the underlying attributes are dichotomous**. Notice that, so far, a third combination, polytomous responses underlain by polytomous attributes, has been rarely discussed in the literature (Chiu et al., 2023; von Davier, 2008) and thus, is not further considered here.

Many of the ideas presented here stem from insights and results obtained while the PI from TC was funded as the recipient of an NSF-CAREER award for the project “*Cognitive Diagnosis in E-Learning: A Nonparametric Approach for Computerized Adaptive Testing*.” That work established the NPCD framework and provided the theoretical basis for the Nonparametric Classification Method (NPC; Chiu & Douglas, 2013; Wang & Douglas, 2015) and the General Nonparametric Classification Method (GNPC; Chiu et al., 2018; Chiu & Köhn, 2019).

The NPCD framework also supported the development of computerized adaptive testing for CD (CD-CAT) and contributed to a refined perspective on Q-matrix (Tatsuoka, 1985) theory. These methodological advances were translated into practice through the development of an interactive application, Computerized Adaptive Testing and Learning for Cognitive Diagnosis (CATL-CD), which enabled CD-CAT in small instructional settings such as classrooms and university courses, where timely formative feedback is essential for monitoring and supporting student learning (De Bra, 2006; GAISE, 2016).

Use of CATL-CD in practice revealed a major limitation of the original NPCD framework: many operational assessments use a polytomous rather than a dichotomous item response format — in not mentioning items underlain by polytomous attributes. Multiple-choice (MC) items are a prominent example. This limitation motivated a substantial extension of NPCD and CATL-CD to accommodate MC items, leading to the development of the nonparametric classification method for MC items (MC-NPC; Wang et al. 2023). The strong performance of MC-NPC in small instructional settings also qualified it to serve as the classification engine for a newly developed CD-CAT algorithm for MC items called MC-NPS (Wang et al. 2025). Together, MC-NPC and MC-NPS constitute the core methodology for CD-CAT with MC items in small educational settings and represented a key step toward broader practical applicability of NPCD methods.

Despite these advances, significant gaps remain. For example, the MC-NPC method treats response options as nominal and is therefore not well suited for items whose response categories have an inherent ordered or sequential structure. Within parametric CD, many of these challenges have been addressed through unifying model frameworks (e.g., the generalized CD model for dichotomous responses underlain by polytomous attributes [sp-CDM] proposed by de la Torre and colleagues, 2025). However, within nonparametric CD, comparable developments are still lacking. In addition, the construction of Q-matrices — particularly in settings involving polytomous attributes — remains labor-intensive and may be prone to human error resulting in misspecifications.

A further limitation arises from the fact that data-driven methods for Q-matrix construction and validation are not applicable in assessment settings with small sample sizes.

Motivated by these difficulties, the present proposal seeks to extend the NPCD framework to polytomous data — specifically, dichotomous item responses underlain by polytomous attributes, or polytomous item responses underlain by dichotomous attributes — in a general and unified manner suitable for small instructional settings. To support the practical implementation of these methods, the project introduces an AI-assisted pipeline for Q-matrix construction that leverages a structured, multi-agent large language model (LLM) framework. This pipeline is designed to be data-free while promoting accessibility, scalability (with respect to the number of items), stability, and computational tractability. It supports the development of initial item–attribute mappings in a transparent and auditable manner by generating evidence-based rationales that facilitate expert review and substantially reduce the time and effort required to develop CD assessments for classroom use.

The next sections summarize key results from prior NSF support and review relevant technical background, followed by a detailed description of the research objectives of the proposed project.

### Results from Prior NSF Support

As part of the mandatory disclosure of previous NSF support, this section briefly summarizes main aspects of the NSF-CAREER award “*Cognitive Diagnosis in E-Learning: A Nonparametric Approach for Computerized Adaptive Testing*,” which provided funding for the PI at TC from 2016 to 2024. These years included three no-cost extensions due to the COVID pandemic and two institutional transfers (# 1552563: Rutgers University; # 2302406: University of Minnesota, Twin Cities; and # 2423762: Teachers College, Columbia University); the total amount was \$731,165.

#### *Intellectual Merit*

The NSF-CAREER grant project made a major contribution in enabling the use of CD-CAT in small-scale educational settings where current implementations fail due to the need for large calibration samples. The NPCD framework provided the theoretical foundation to resolve the sample size issue. Core components of the nonparametric CD-CAT are the Nonparametric Item Selection (NPS) method (Chang et al. 2019) for DINA data, and for general CDMs the General Nonparametric Item Selection (GNPS) method (Chiu & Chang, 2021) that control the flow of items as they are chosen to match an examinee’s intermediate ability assessment. GNPC is the computational engine for steadily updating the assessment of an examinee’s ability.

#### *Broader Impacts*

Many experts in the field predict that CD is the future of educational measurement (Huff & Goodman, 2007; Stout, 2002). Compared with traditional psychometric approaches, CD allows for a finer-grained assessment of an examinee’s mental ability than can be obtained through single-score models. The advent of CAT marked a huge breakthrough for educational testing because CAT allows for shorter, less time-consuming tests that are more accurate than paper-pencil tests. Thus, CD-CAT combines the best of two worlds, as it uses personalized assessments, which generate accurate and detailed diagnostic information on a student’s mental ability.

For large-scale testing, CD-CAT is well-developed; fast and efficient likelihood-based estimation routines are available. However, CD-CAT becomes computationally infeasible when sample sizes are small. Hence, CD-CAT is typically not available where it might be really useful: in the classroom, in university courses — when learning is actually happening, and teachers and students need it most. In summary, the contribution of the nonparametric CD-CAT system as it was developed with the NSF-CAREER award cannot be overstated. The CATL-CD application was used with

courses the PI taught at Rutgers, The State University of New Jersey. CATL-CD was accessible by smart phone and tablet so that student could practice whenever time permitted. Finally, one should notice that CATL-CD is not limited to the realm of statistics education, but can readily be applied to any discipline.

The NSF-CAREER project has provided research opportunities and financial support for five Ph.D. students; three of the students have published papers with the PI in highly prestigious outlets like *Psychometrika* and *Applied Psychological Measurement*.

### *Research Products*

The NSF-CAREER project has produced 10 publications in highly prestigious outlets such as *Psychometrika*, *Applied Psychological Measurement* and the *Journal of Classification*: (1) Oh & Chiu (online first); (2) Köhn, Chiu, Oluwalana, Kim & Wang (2024); (3) Chiu & Chang (2021); (4) Köhn & Chiu (2021); (5) Chiu & Köhn (2019a); (6) Chiu & Köhn (2019b); (7) Chang, Chiu, & Tsai (2019); (8) Köhn & Chiu (2019); (9) Köhn & Chiu (2018); (10) Chiu, Sun & Bian (2018). The research has spawned many conference presentations and workshops. Five out of the 10 papers are accomplished with 7 students and almost all presentations are collaborations with students supported by the grant. The two newly developed CD-CAT algorithms are included in the `cdcatR` R package; the other R routines for NPCD are available in the `NPCDTools` package. The PI maintains a website that provides researchers in education, measurement, and statistics with detailed information and updates on the methodological developments and progress of the project.

### *Further Insights and Learnings*

The applicability of CATL-CD is supported through interdisciplinary collaborations. For example, the PI and colleagues received Faculty Collaboration Planning Funds at Teachers College, Columbia University, to develop an item pool for mapping human potential for transformative learning using the CATL-CD platform. Feedback confirms the advantages of methods that can be used to analyze polytomous items and encourages the continuing development of CATL-CD for polytomous data.

## Technical Background

### *Cognitive Diagnosis: A Brief Review*

*Skills, Attributes, and Proficiency.* CDMs describe ability in a given knowledge domain as a composite of  $K$  cognitive skills—henceforth: “attributes”—that a student has mastered or not (DiBello et al., 2007; Haberman & von Davier, 2007; Leighton & Gierl, 2007; Nichols et al., 1995; Rupp et al., 2010; Sessoms & Henson, 2018; Tatsuoka, 2009). Attribute mastery is recorded as a  $K$ -dimensional binary vector  $\alpha = (\alpha_1, \dots, \alpha_K)'$ . Different zero-one combinations define attribute profiles characterizing distinct classes of ability or proficiency,  $\mathcal{C}_m$ ,  $m = 1, 2, \dots, M = 2^K$ . (The terms profile and vector are used interchangeably here.) The primary task of CD is to assign students to one of these  $M$  classes based on their performance in a test that targets proficiency in the knowledge domain in question. Said differently, examinees’ individual attribute vectors  $\alpha_i$  must be estimated ( $i = 1, \dots, N$  is the examinee index).

*Q-Matrix Essentials.* Like examinees, so are the items of CD assessments characterized by individual profiles that determine which specific attributes are required to respond correctly to an item. Formally, the item-attribute profile is a  $K$ -dimensional binary vector denoted as  $\mathbf{q}_j$ , with elements  $q_{jk} = 1$  if a correct answer requires mastery of the  $k^{th}$  attribute  $\alpha_k$ , and 0 otherwise. Given  $K$  attributes, at most  $2^K - 1$  distinct item-attribute profiles can be constructed. The  $J$  item-attribute profiles of a given test are collected into a  $J \times K$  matrix called the Q-matrix,  $\mathbf{Q} = \{q_{jk}\}_{(J \times K)}$  (Tatsuoka, 1985). The Q-matrix documents the associations between items and attributes. The correct specification of the Q-matrix is mandatory for the validity of CD. An

incorrect Q-matrix causes examinees to be assigned to proficiency classes to which they do not belong regardless of which specific CDM is chosen for modeling the data (e.g., Chiu, 2013; de la Torre & Chiu, 2016). As the key condition for the identifiability of any CDM, the Q-matrix must possess the completeness property. Loosely speaking, completeness means that the Q-matrix is composed of item attribute vectors that allow for identifying all proficiency classes (Chiu et al., 2009; Köhn & Chiu, 2017, 2018, 2019, 2021).

#### *The Nonparametric Classification (NPC) Method*

The NPC method developed by Chiu and Douglas (2013) does not rely on parametric estimation of examinees’ proficiency class membership, but uses a distance-based algorithm on the observed item responses for classifying examinees. Proficiency class membership is determined by comparing an examinee’s observed item response vector  $\mathbf{Y}$  with each of the ideal item response vectors of the realizable  $M$  proficiency classes. The ideal item responses are a function of the Q-matrix and the attribute vectors characteristic of the different proficiency classes. Hence, an examinee’s proficiency class is identified by the attribute vector  $\alpha_m$  underlying that ideal item response vector which is closest—or most similar—to an examinee’s observed item response vector.

#### *The General Nonparametric Classification Method (GNPC)*

The consistency conditions of the NPC-estimator  $\hat{\alpha}$  identified by Wang and Douglas (2015) are often difficult to meet for more complex CDMs because the probability of a correct response to an item increases as a function of the number of required attributes that are mastered by an examinee (known as the “monotonicity assumption”). Hence,  $\eta$ , as it was used with the NPC method, does not offer the necessary flexibility to model the relation between required and mastered attributes for these complex CDMs.

Define the conjunctive ideal response  $\eta_{ij}^{(c)} = \prod_{k=1}^K \alpha_{ik}^{q_{jk}}$  and the disjunctive ideal response  $\eta_{ij}^{(d)} = 1 - \prod_{k=1}^K (1 - \alpha_{ik})^{q_{jk}}$ . Chiu et al. (2018) proposed a weighted ideal response  $\eta^{(w)}$ , defined as the convex combination of  $\eta^{(c)}$  and  $\eta^{(d)}$ , for the development of the General NPC (GNPC) method. For each item  $j$  and proficiency class  $\mathcal{C}_m$ , the weighted ideal response  $\eta_{mj}^{(w)}$  is defined as the convex combination

$$\eta_{mj}^{(w)} = w_{mj}\eta_{mj}^{(d)} + (1 - w_{mj})\eta_{mj}^{(c)}, \quad (1)$$

where  $0 \leq w_{mj} \leq 1$ . As an important feature of the GNPC method, the weights  $w_{mj}$  are estimated from the data and do not require a priori knowledge of the CDM underlying the data; thus,  $w_{mj}$  is “automatically” adjusted to the level of complexity and variability of the data. The weight  $w_{mj}$  can be estimated by minimizing the distance between the observed responses to item  $j$  and the weighted ideal responses  $\eta_{mj}^{(w)}$  of examinees in  $\mathcal{C}_m$ . Subsequently,  $\hat{\eta}_{mj}^{(w)}$  is computed from  $\hat{w}_{mj}$  based on Equation (1). The GNPC estimator  $\hat{\alpha}$  of an examinee’s attribute vector is defined as the attribute vector underlying the weighted ideal item response vector that among all weighted ideal item response vectors minimizes the distance between the observed item response vector and a particular weighted ideal item response vector  $\eta_m^{(w)}$ . The consistency of the GNPC estimator was proven by Chiu and Köhn (2019b).

#### *Parametric Cognitive Diagnosis for Polytomous Data*

As previously mentioned, parametric approaches to CD for handling polytomous data can be grouped into three categories.

The first category consists of CDMs for dichotomous responses that are presumed to be underlain by polytomous attributes. Early work in this area includes the reparameterized unified model (Hartz, 2002) and Karelitz’ (2004) ordered-category attribute coding (OCAC) framework, which

was developed to represent multiple levels of ordered attributes. Karelitz (2004) implemented the OCAC system within the Deterministic Input, Noisy “AND” gate (DINA) model (e.g., Junker & Sijtsma, 2001). The OCAC framework was subsequently adopted in polytomous-attribute extensions of both conjunctive and compensatory CDMs, in which item success probabilities increase monotonically with higher levels of attribute mastery. For example, Chen and de la Torre (2013) proposed an extension of the Generalized DINA (G-DINA; de la Torre, 2011) model, the polytomous G-DINA (pG-DINA) model, designed to accommodate polytomous attributes. However, the computationally expensive calibration of the pG-DINA remained a major obstacle to its application in practice (Chen & de la Torre, 2013; Ma, 2022). In an effort to further reduce model complexity, Ma (2022) proposed the higher-order general cognitive diagnosis model with polytomous attributes for dichotomous responses (HO-PCDM). This model employs a sequential attribute-level recoding procedure that integrates Karelitz’s (2004) OCAC paradigm with the concept of pseudo-attributes introduced by Bao (2019). Despite its reformulation as a higher-order model (de la Torre & Douglas, 2004), the HO-PCDM still requires relatively large sample sizes. Extensions of broader diagnostic frameworks, including the general diagnostic model (GDM; von Davier, 2008) and the log-linear cognitive diagnosis model (Henson et al., 2009), have also been developed to accommodate ordinal attributes while modeling dichotomous item responses.

The second category comprises CDMs for polytomous responses underlain by dichotomous attributes (e.g., CDMs that accommodate ordered or graded response processes through partial-credit, sequential, or rating-scale mechanisms). Early work in this area includes the GDM (von Davier, 2008), which accommodates polytomous responses within a diagnostic framework while maintaining dichotomous latent attributes. Subsequent models introduced more restrictive but cognitively interpretable structures, such as the partial-credit DINA model (de la Torre, 2010), the polytomous log-linear cognitive diagnosis model (Hansen, 2013) and the sequential G-DINA model (Ma & de la Torre, 2016), each relying on different assumptions about the response process (e.g., graded, partial-credit, or sequential mechanisms). More recent developments, including rating-scale diagnostic models (Liu & Jiang, 2020), further expanded the applicability of CDMs to ordered categorical data. Within this line of research, Gao et al. (2021) proposed a general class of CDMs for polytomous responses that unifies these existing models under a common framework. By allowing different link functions (cumulative, adjacent-category, and continuation-ratio logits), category-level Q-matrices and parameter constraints across response categories, their approach subsumes many existing polytomous CDMs as special cases and offers greater modeling flexibility. As such, the framework of Gao et al. (2021) represents an important step toward a systematic and coherent treatment of polytomous response processes in CD assessment.

The third category consists of CDMs with both polytomous responses and polytomous attributes. GDM is the earliest model that can deal with such data. The polytomous DINA model (Chiu et al., 2023) is another more recent model that generalizes the DINA model to settings in which both latent attributes and item responses are polytomous.

## Research Plan 1: Methodological and Computational Developments

### *Objective 1: Development of the Unified NPCD Framework for Polytomous Data*

In practice, assessments with polytomous structures are abundant. Response formats are often polytomous such as MC, partial-credit scoring, rubric-based evaluations, and strategy or sequential step/process selections. At the attribute level, cognitive skills are often more realistically represented as multi-level or developmental constructs (e.g., emerging, developing, proficient) rather than plain (non-)mastered states. Collapsing either responses or attributes into a dichotomous format risks discarding diagnostically relevant information and undermines the differentiated assess-

ment that is intended by CD. The original NPCD framework cannot accommodate these complex structures. Hence, the first objective of the project proposed here is the development of a unified nonparametric classification framework for polytomous data (p-GNPC).

The key idea is to take a parametric CDM designed for polytomous data — whether polytomous responses under dichotomous attributes (e.g., sequential/graded response processes) or dichotomous responses under polytomous attributes (e.g., multi-level skill requirements) — and **re-express it in the general form required by the GNPC method thereby transitioning from parametric to nonparametric CD**. Concretely, this is accomplished by (i) recoding polytomous structures into a set of binary indicators (e.g., category-threshold indicators or level-indicator attributes) and (ii) defining the corresponding ideal responses so that GNPC’s weighted ideal-response and distance-based classification machinery can be used without resorting to a particular link function or parameterization of item responses. The following two examples illustrate this “re-cast into GNPC” principle in both directions.

*Example 1: Polytomous responses with dichotomous attributes (sequential G-DINA model).*

The first example demonstrates a p-GNPC construction for data conforming to the sequential G-DINA model (Ma & de la Torre, 2016), which accommodates polytomous responses while maintaining dichotomous attributes under a sequential response process. Consider the following item that concerns the use of carbon footprint as key indicator to represent contributions to global warming:

School bus service shows a global warming potential (GWP) of 60 kg CO<sub>2-eq</sub>/day. It is known that you ride the bus for 5 miles per day. Here is the daily travel pattern and passenger profile.

| Travel Distance (Miles) | Number of Passengers (Persons) |
|-------------------------|--------------------------------|
| 5                       | 50                             |
| 10                      | 20                             |
| 15                      | 10                             |

For how much GWP are you personally responsible for in terms of your commute?

To solve this problem, an examinee is supposed to master several sequential steps and the corresponding attributes in the category-level  $q$ -vectors:

| Step                                                              | Response Category | Attributes |            |            |
|-------------------------------------------------------------------|-------------------|------------|------------|------------|
|                                                                   |                   | $\alpha_1$ | $\alpha_2$ | $\alpha_3$ |
| Step 1: Compute $5 \times 50$ , $10 \times 20$ and $15 \times 10$ | 1                 | 1          | 0          | 0          |
| Step 2: $5 \times 50 + 10 \times 20 + 15 \times 10 = 600$         | 2                 | 0          | 1          | 0          |
| Step 3: $\frac{1 \times 5}{600} \times 60 = 0.5$                  | 3                 | 0          | 0          | 1          |

The three required attributes are

$\alpha_1$ : Quantifying environmental impact by applying a feasible functional unit.

$\alpha_2$ : Calculating the total service provided.

$\alpha_3$ : Determining the shared environmental responsibility of the target consumer.

Let  $X_j$  denote the polytomous response with categories  $t = 1, \dots, T$  (with  $t$  indexing the sequential step/category). Denote the category-level  $q$ -vector for item  $j$  in category  $t$  as  $\mathbf{q}_{jt}$ , where  $j = 1, \dots, J$ . The item-level  $q$ -vector can then be obtained by  $\mathbf{q}_j = \bigvee_{u=1}^T \mathbf{q}_{ju}$  and thus, the number of required attributes for item  $j$  is  $K_j^* = \sum_{k=1}^K q_{jk}$ . For notational convenience, reorder the attributes in  $\mathbf{q}_j$  and  $\boldsymbol{\alpha}_m$  so that the  $K_j^*$  required attributes appear in the first  $K_j^*$  positions, and write

$\alpha_m^* = (\alpha_{m1}, \alpha_{m2}, \dots, \alpha_{mK_j^*})$ ,  $m = 1, \dots, 2^{K_j^*}$ . Define  $\mathbf{q}_{jt}^+ = \bigvee_{u \leq t} \mathbf{q}_{ju}$  the q-vector required to advance to category  $t$  or above. Then for item  $j$ , the number of required attributes to achieve category  $t$  or above is  $K_{jt}^{*+} = \sum_{k=1}^K q_{jtk}^+$ . Gao and colleagues (2021) showed that the sequential G-DINA model is equivalent to the general class of CDMs for polytomous responses with continuation-ratio logit link, expressed as:

$$\log \left( \frac{P(X_j \geq t \mid \alpha_m^*)}{P(X_j = t-1 \mid \alpha_m^*)} \right) = \beta_{jt0} + \sum_{k=1}^{K_{jt}^{*+}} \beta_{jtk} \alpha_{mk} + \sum_{k=1}^{K_{jt}^{*+}-1} \sum_{k'=k+1}^{K_{jt}^{*+}} \beta_{jtkk'} \alpha_{mk} \alpha_{mk'} + \dots + \beta_{jt12\dots K_{jt}^{*+}} \prod_{k=1}^{K_{jt}^{*+}} \alpha_{mk}. \quad (2)$$

Now recode the data category-by-category: for each  $t$ , define a binary indicator

$$Y_{jt} = \begin{cases} 1 & \text{if } X_j \geq t, \\ 0 & \text{if } X_j = t-1. \end{cases}$$

Then Equation (2) can be rewritten as a dichotomous-response G-DINA model:

$$\log \left( \frac{P(Y_{jt} = 1 \mid \alpha_m^*)}{P(Y_{jt} = 0 \mid \alpha_m^*)} \right) = \beta_{jt0} + \sum_{k=1}^{K_{jt}^{*+}} \beta_{jtk} \alpha_{mk} + \sum_{k=1}^{K_{jt}^{*+}-1} \sum_{k'=k+1}^{K_{jt}^{*+}} \beta_{jtkk'} \alpha_{mk} \alpha_{mk'} + \dots + \beta_{jt12\dots K_{jt}^{*+}} \prod_{k=1}^{K_{jt}^{*+}} \alpha_{mk}.$$

Thus, a polytomous item under a sequential model can be treated as a collection of dichotomous sub-items indexed by  $t$ . This is precisely the GNPC “entry point”: once responses are recoded into  $Y_{jt}$ , the category-level GNPC method can be applied to each  $(j, t+)$  dichotomous subproblem to construct an “aggregated” weighted ideal response. After the “aggregated” weighted ideal responses across all  $t$  are computed, the individual weighted ideal response at category  $(j, t)$  can be obtained by taking the difference between the aggregated ones for  $(j, t+)$  and  $(j, (t+1)+)$ . Ultimately, the p-GNPC classification for the original polytomous item can be obtained. For clarity, quantities derived using data  $Y$  are denoted with a superscript “+” to distinguish them from those associated with  $X$ . The next paragraph describes the concept in detail.

For each latent class  $m$ , where  $m = 1, \dots, 2^{K_j^*}$ , define the “aggregated” conjunctive and disjunctive ideal responses as,

$$\eta_{mjt}^{(c+)} = \prod_{k=1}^{K_{jt}^{*+}} \alpha_{mk}^{q_{jtk}^+} \quad \text{and} \quad \eta_{mjt}^{(d+)} = 1 - \prod_{k=1}^{K_{jt}^{*+}} (1 - \alpha_{mk})^{q_{jtk}^+}.$$

Similar to Equation (1), the aggregated weighted ideal response  $\eta_{mjt}^{(w+)}$  can be formulated as

$$\eta_{mjt}^{(w+)} = w_{mjt}^+ \eta_{mjt}^{(d+)} + (1 - w_{mjt}^+) \eta_{mjt}^{(c+)}, \quad (3)$$

where  $w_{mjt}^+$  controls how close  $\eta_{mjt}^{(w+)}$  is to  $\eta_{mjt}^{(d+)}$ . Using the distance between observed responses and weighted ideal responses,

$$d_{mjt}^+ = \sum_{i \in C_m} (Y_{ijt} - \eta_{mjt}^{(w+)})^2,$$

where  $C_m$  is the latent class corresponding to  $\alpha_l^*$ , the weight for the case  $\eta_{mjt}^{(c+)} = 0$  and  $\eta_{mjt}^{(d+)} = 1$  is obtained by minimizing  $d_{mjt}^+$ , resulting in the closed form:

$$\hat{w}_{mjt}^+ = \frac{\sum_{i \in C_m} (y_{ijt} - \eta_{mjt}^{(c+)})}{\|C_l\|(\eta_{mjt}^{(d+)} - \eta_{mjt}^{(c+)})} = \bar{y}_{mjt} \quad (4)$$

Plugging Equation (4) into Equation (3) gives  $\hat{\eta}_{mjt}^{(w+)} = \bar{y}_{mjt}$  as well. Notice that when  $\eta_{mjt}^{(c+)} = \eta_{mjt}^{(d+)} = 0$  and  $\eta_{mjt}^{(c+)} = \eta_{mjt}^{(d+)} = 1$ ,  $\eta_{mjt}^{(w+)}$  is fixed at 0 and 1, respectively, and no weight estimation is needed. After all  $\eta_{mjt}^{(w+)}$  have been estimated,  $\eta_{mjt}^{(w)}$  for a particular category  $t$  can be obtained by computing

$$\eta_{mjt}^{(w)} = \eta_{mjt}^{(w+)} - \eta_{mj(t+1)}^{(w+)}.$$

Notice that  $\eta_{mjt}^{(w)} \geq 0$  because the item is sequential in nature, and thus, the number of examinees in the group of  $Y_{jt} = 1$  is greater than or equal to that in the group of  $Y_{j(t+1)} = 1$ . In other words,  $\bar{y}_{mjt} \geq \bar{y}_{mj(t+1)}$  due to  $X_{jt} \geq 1$ , implying that  $\eta_{mjt}^{(w+)} \geq \eta_{mj(t+1)}^{(w+)}$ .  $\hat{\alpha}_i$  is then obtained by minimizing the distance between  $X_{ijt}$  and  $\eta_{mjt}^{(w)}$  over all  $t$  and  $j$ :

$$d(\mathbf{x}_i, \hat{\eta}_m^{(w)}) = \sum_{j=1}^J \sum_{t=1}^T (x_{ijt} - \hat{\eta}_{mjt}^{(w)})^2 \Rightarrow \hat{\alpha}_i = \arg \min_{\alpha_m} d(\mathbf{x}_i, \hat{\eta}_m^{(w)})$$

with  $m \in \{1, \dots, 2^{K_{jt}^{*+}}\}$ . The key idea implemented in the p-GNPC framework is nicely illustrated by this example: The response-category probabilities of the parametric sequential G-DINA model are replaced by weighted ideal responses each defined for different polytomous levels. Said differently, within the p-GNPC, polytomous response structures are re-expressed as a collection of dichotomous components, the ideal responses. Examinees are then classified through distance minimization.

*Example 2: Dichotomous responses with polytomous attributes (sp-CDM)*

The second example illustrates the complementary case: dichotomous responses under polytomous attributes. Consider the following example used in Tjoe and de la Torre (2014):

Danny's 4-in. by 6-in. photo of his skateboard is enlarged into a 12-in. by 18-in. photo. If the length of his skateboard in the original photo is 5 in. how long is his skateboard in the enlarged photo?

(A) 11 in. (B) 13 in. (C) 15 in. (D) 17 in.

The key of the item is (C), and four attributes are required:

- $\alpha_1$ : Prerequisite skills and concepts required in proportional reasoning (multiplication skills)
- $\alpha_3(b)$ : Constructing proportions from a given situation
- $\alpha_4$ : Identifying a multiplicative relationship between sets of values
- $\alpha_5$ : Determining whether two relations form a proportion

The other two attributes,  $\alpha_2$ : "comparing and ordering fractions" and  $\alpha_6$ : "applying algorithms in solving proportional reasoning problems," are not required. Notice that  $\alpha_3(b)$  is the second level of the broader attribute  $\alpha_3$ , with  $\alpha_3(a)$ : "constructing ratios from a given situation" representing the first level. This item can therefore be analyzed with dichotomous responses and polytomous attributes. As discussed earlier, the sp-CDM (de la Torre et al., 2025) provides a general parametric model for such data. Without loss of generality, let  $M$  be the number of levels for each attribute (the discussion extends directly to attribute-specific  $M_k$ ). In this example, the number of required attributes for item  $j$  is  $K_j^* = \sum_{k=1}^K I[q_{jk} > 0]$ . We write  $\alpha_l^* = (\alpha_{l1}, \alpha_{l2}, \dots, \alpha_{lK_j^*})$ ,  $l = 1, \dots, M^{K_j^*}$  to distinguish from  $\alpha_m^*$  used in Example 1 to denote dichotomous attribute profile. Using the

identity link, the saturated sp-CDM item response function can be expressed as

$$\begin{aligned}
P(Y_j = 1 | \boldsymbol{\alpha}_l^*) &= \delta_{j0} + \sum_{k=1}^{K_j^*} \sum_{m=1}^{M-1} \delta_{jkm} I[\alpha_{lk} = m] + \sum_{k=1}^{K_j^*-1} \sum_{k'=k+1}^{K_j^*} \sum_{m=1}^{M-1} \sum_{m'=1}^{M-1} \delta_{jkmk'm'} I[\alpha_{lk} = m] I[\alpha_{lk'} = m'] \\
&\quad + \cdots + \sum_{m_1=1}^{M-1} \cdots \sum_{m_{K_j^*}=1}^{M-1} \delta_{j1m_1 2m_2 \dots K_j^* m_{K_j^*}} \prod_{k=1}^{K_j^*} I[\alpha_{lk} = m_k]
\end{aligned}$$

The p-GNPC counterpart is obtained by re-casting the model in GNPC form via ideal responses defined on polytomous attribute levels. For each latent class  $l, l = 1, \dots, M^{K_j^*}$  and item  $j, j = 1, \dots, J$ , define the conjunctive and disjunctive ideal responses as

$$\eta_{lj}^{(c)} = \begin{cases} 1 & \text{if } \prod_{k=1}^{K_j^*} I[\alpha_{lk} \geq q_{jk}] = 1, \\ 0 & \text{otherwise} \end{cases}, \quad \text{and} \quad \eta_{lj}^{(d)} = \begin{cases} 0 & \text{if } \prod_{k=1}^{K_j^*} (1 - I[\alpha_{lk} \geq q_{jk}]) = 1, \\ 1 & \text{otherwise.} \end{cases}$$

Analogous to GNPC, a weighted ideal response is then defined using Equation (1) and  $\hat{\boldsymbol{\alpha}}_i$  can then be obtained by minimizing the distance between the observed item response pattern  $\mathbf{y}_i$  and the weighted ideal response pattern  $\hat{\boldsymbol{\eta}}_i^{(w)}$ . The detail is omitted here.

In conclusion, the unified approach to nonparametric polytomous CD implemented in the p-GNPC supports stable and consistent classification, avoids computational issues that parametric methods may encounter in small-scale educational settings (Wang et al., 2023), and produces rich and accurate diagnostic evidence (e.g., partial credit and developmental progressions) without sacrificing interpretability.

Future development will focus on extending the p-GNPC framework to incorporate nonparametric adaptation of a broader class of polytomous CDMs, including models corresponding to the cumulative and adjacent-category link functions in the general class of polytomous models (Gao et al., 2021), as well as other related variants. Ultimately, integrating these diverse response processes will establish p-GNPC as a comprehensive and unified environment for CD, capable of accommodating complex measurement structures within a single hub designed for small-scale assessment.

### *Objective 2: AI-Assisted Q-Matrix Construction for Polytomous CD*

Constructing high-quality Q-matrices remains a major challenge in CD, as the true Q-matrix underlying an assessment is never directly observed. In practical, Q-matrices are typically developed through expert workshops followed by iterative revisions. Such a process is costly, may be inconsistent across experts and administrations, and is error-prone — especially, when the attributes are polytomous and their number is large. A misspecified Q-matrix can distort item parameter estimates and may lead to incorrect classification of examinees.

To address these limitations, data-driven Q-matrix estimation and validation procedures for polytomous data have been proposed (de la Torre et al., 2022; He et al., 2023; Ma & de la Torre, 2020). However, such approaches generally require large sample sizes and are therefore not well suited to settings in which response data are limited or unavailable. As a result, constructing valid Q-matrices — particularly for polytomous data — remains difficult for assessments intended to support teaching and learning in small-scale instructional contexts.

To mitigate the limitations of expert-only Q-matrix construction while avoiding dependence on response data, we propose an automated, auditable, and data-independent procedure based on a structured multi-agent LLM pipeline, in which specialized AI modules (hereafter, “agents”) perform distinct subtasks. The pipeline decomposes Q-matrix construction into explicit, traceable stages,

allowing each q-entry to be linked to supporting evidence and, when relevant, to disagreement signals. A central design feature is the use of agent committees — multiple independent agents performing the same task in parallel — whose outputs are consolidated through consensus. This design is termed *Traceable Agent-Committee Encoding for Q-matrix construction* (TRACE-Q). Unlike supervised text-classification approaches that rely on response data and pre-determined skill labels (e.g., Zhao & Huang, 2020; Pardos & Dadu, 2017), TRACE-Q constructs a Q-matrix directly from raw item statements without requiring any data, making it suitable for new assessments or small-sample settings. Compared to other text-only AI/ML approaches (e.g., Chen et al., 2025), it offers a more transparent and traceable construction process. Although fully automated, all outcomes are documented with supporting evidence, enabling efficient and targeted human review.

Table 1 illustrates the role of each agent in the system. TRACE-Q is implemented in `Python` using the `openai` and `pydantic` libraries. In the current implementation, TRACE-Q uses GPT-5.2 via OpenAI Chat Completions API with `temperature = 0`, `seed = 42`, and structured JSON outputs. Other generation parameters (e.g., `top_p`, `max_tokens`, and stop sequences) use provider defaults. The pipeline proceeds in four stages: Solution Generation (Steps 1-2) produces verified solution keys; Skill Extraction (Steps 3-4) defines a unified skill codebook; Skill Annotation (Steps 5-6) assigns q-vectors to each item; and Review (Step 7) audits the final output. Figure 1 illus-

Table 1: Overview Workflow of TRACE-Q (Icons from Flaticon <https://www.flaticon.com>).

| Stage               | LLM Agent                                                                                                    | Functionality                                                                                                                                                                                                                                                                                                     |
|---------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Solution Generation |  1. Solver                  | Generated step-by-step solution key for each item, where each solution step must show explicit reasoning.                                                                                                                                                                                                         |
|                     |  2. Verifier               | Independently resolves all the items and compares the results against the Solver's output. If discrepancies are detected, it corrects the solutions and produces verified solutions.                                                                                                                              |
| Skill Extraction    |  3. Expert ( $\times 3$ ) | Three independent agents separately analyze the verified solutions and each proposes a skill codebook. Skills must be aligned with the solution steps, cognitively distinct, and at appropriate granularity (typically 4-8 skills). Each skill includes a definition, inclusion/exclusion criteria, and examples. |
|                     |  4. Supervisor            | Consolidates the three Experts' skill codebooks into one unified codebook. It aligns equivalent skills across experts (even if named differently), prioritizes consensus (3/3 = high confidence; 2/3 = include; 1/3 = evaluate carefully), and resolves conflicts by merging, renaming, or removing skills.       |
| Skill Annotation    |  5. Tagger ( $\times 5$ ) | Each of the five independent agents assigns a minimal set of required skills based on the final skill codebook. Each tagger must cite step-level evidence from the verified solutions and explain reasoning for inclusion.                                                                                        |
|                     |  6. Judge                 | Aggregates the q-vectors from the Taggers. If agreement is strong ( $\geq 4/5$ ), it accepts the consensus; otherwise, it adjudicates by evaluating each Tagger's work, and outputs the final q-vectors with confidence scores.                                                                                   |
| Review              |  7. Auditor               | Performs a final quality review on each item's q-vector by comparing it with the outcomes from the previous agents. It assigns a verdict (correct/error/controversial) and proposes corrections with justification. Controversial cases are flagged for potential human reviews.                                  |

trates the workflow of the current TRACE-Q pipeline for constructing a provisional Q-matrix for Tatsuoka's (1990) fraction-subtraction assessment (dichotomous attributes and dichotomous responses) using Item 5: " $4 \frac{3}{5} - 3 \frac{4}{10} = ?$ " as an example. The resulting skill set identifies four core attributes: mixed number conversion (**S01**); finding common denominators (**S02**); performing subtraction with compatible forms (**S03**); and simplifying fractions (**S04**).

A preliminary study was conducted to evaluate the performance of the current TRACE-Q against two benchmarks: (a) an expert-constructed Q-matrix with eight attributes and (b) a Q-

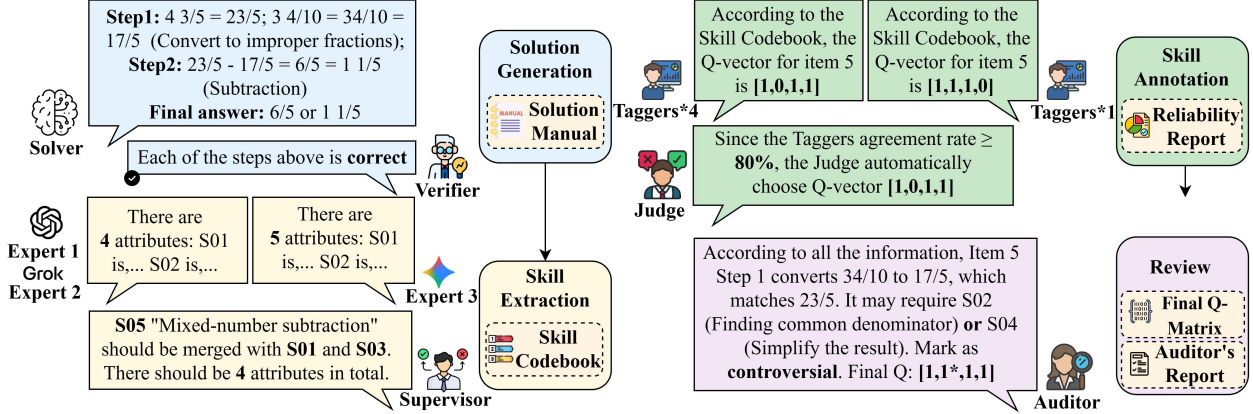


Figure 1: TRACE-Q Workflow Using Item 5: “4 3/5 - 3 4/10 = ?” for Illustration.

matrix estimated by the Two-Step Q-matrix Estimation (TSQE) method (Köhn et al., 2024) with four attributes. Model fit was assessed by fitting the data with the G-DINA model; relative and absolute fit indices are reported in Table 2. Results indicate that TRACE-Q achieved the lowest

Table 2: Model Fit Comparison Across Q-Matrices Obtained from Difference Methods.

| Q-Matrix            | Relative |          |          |         | Absolute           |        |
|---------------------|----------|----------|----------|---------|--------------------|--------|
|                     | AIC      | BIC      | CAIC     | SABIC   | RMSEA <sub>2</sub> | SRMSR  |
| Expert ( $K = 8$ )  | 9421.63  | 11328.07 | 11773.07 | 9915.49 | —                  | 0.0706 |
| TSQE ( $K = 4$ )    | 8961.56  | 9505.64  | 9632.64  | 9102.50 | 0.0423             | 0.0660 |
| TRACE-Q ( $K = 4$ ) | 8843.12  | 9430.04  | 9567.04  | 8995.16 | 0.0369             | 0.0403 |

**Note 1.** RMSEA<sub>2</sub> is unavailable for Expert because  $M_2$  could not be computed due to low degrees of freedom.

**Note 2.** The Q-matrix constructed here by TRACE-Q did not go through expert review.

values across all fit indices, suggesting improved fit relative to the expert and TSQE Q-matrices.

Similar to other data-driven procedures, TRACE-Q concludes with an expert consolidation phase, where the AI-generated provisional Q-matrix is reviewed and revised, if necessary, by human content experts to ensure validity. Such a hybrid approach is mandatory because existing Q-matrix validation methods for polytomous data — specifically Ma and de la Torre (2020) for polytomous responses and de la Torre et al. (2022) for polytomous attributes — rely on item parameter estimates and large sample sizes that are unavailable within the NPCD framework.

The TRACE-Q algorithm described here is preliminary but promising. Several developments are required to further verify and consolidate its performance. First, extension to polytomous data can be achieved by modifying and adding agents and revising the prompts to accommodate graded attribute mastery and/or partial-credit scoring. Second, accuracy, stability, and failure modes should be evaluated through extensive simulation studies (See “Objective 4(a)” for details).

## Research Plan 2: Evaluation of the Success of the Proposed Methods

### Objective 3: Theoretical Evaluation

We believe that the theoretical justification for a newly developed method cannot rest solely on “plausibility,” but must instead be established through rigorous mathematical derivations or formal proofs (e.g., Chiu & Köhn, 2019b; Chiu et al., 2018; Köhn & Chiu, 2017; Köhn & Chiu, 2019; Wang et al., 2023). For the same reason, a theoretical justification is required for the proposed unified p-GNPC framework. Specifically, the estimators of examinee proficiency classes must be shown to be statistically consistent. Also, the rates of convergence of the methods need to be investigated,

as they are critical indicators of the efficiency and effectiveness of any computational algorithm.

*Objective 4(a): Empirical Evaluation - Simulations*

*For the p-GNPC methods.* Another key question concerns the efficiency and effectiveness of the proposed p-GNPC methods in comparison with model-based approaches that rely on parametric estimation procedures. This important technical issue pertains to both computational performance and classification accuracy — specifically, the accuracy with which examinees are assigned to their true proficiency classes. Parametric methods can be expected to perform well when large samples of examinees are available and when the CDM underlying the observed data is correctly identified. If these conditions are violated, then the performance of parametric methods is likely to deteriorate. In contrast, nonparametric methods are insensitive to sample size restrictions and misspecification of the CDM underlying the data. These claims are examined through extensive simulation studies that systematically vary (a) the sample size and (b) the choices of CDMs underlying the data. Because the data for these simulations are generated synthetically, the true proficiency class of each student is known. Hence, the rate of correct classification provides a direct and informative basis for comparing the performance of both approaches.

*For TRACE-Q.* Simulations will be used to evaluate the internal stability of TRACE-Q and its external agreement with established Q-matrix estimation and validation methods. Internal stability will be assessed by examining TRACE-Q outputs under controlled perturbations of (a) item characteristics (e.g., linguistic complexity, response format), (b) content domains (e.g., mathematics, natural sciences, or humanities), and (c) pipeline design (e.g., choices of LLM and prompt variants). Stability will further be assessed by removing or simplifying individual pipeline modules and by varying committee size, allowing estimation of each module’s contribution to the resulting Q-matrix. Stability will be measured using inter-replication agreement rates and uncertainty indices reflecting variability in attribute assignment. External agreement will be evaluated by comparing TRACE-Q against established data-driven methods for polytomous data (e.g., de la Torre et al., 2022; He et al., 2023; Ma & de la Torre, 2020), as well as by examining downstream diagnostic performance under NPCD-based methods. To mitigate LLM stochasticity, variability across runs will be explicitly recorded to ensure reproducibility and interpretability of the proposed TRACE-Q.

*Objective 4(b): Empirical Evaluation - Real Data Analysis*

Simulations are an important tool for evaluating new methods, but they provide only a partial picture because the data are generated under well-defined and controlled conditions. Real-world data, by contrast, often deviate from such regularity and predictability. Consequently, best research practice calls for evaluating the performance of the proposed methods using empirical data as well.

Although polytomous real-world datasets have been used in prior studies, these data are either not publicly available or were not collected with CD-based assessments. Therefore, we will collect original data using assessments specifically designed for this project. The newly developed items will be used to assess student learning in a master-level course titled “*HUMD 5122: Applied Regression Analysis*”, a multi-section course offered by the Department of Human Development at TC. The course textbook is “*Multiple Regression: A Primer*” (Allison, 1998). The PI from TC teaches one of the three sections and will work closely with the other instructors to select items that align with the course curriculum and reflect the learning goals articulated in the GAISE (2016) guidelines.

The PI is an experienced developer of CD assessments. In her NSF CAREER project, she and her PhD student developed an item bank of more than 150 items, each with associated q-vectors, using Jackson and Macpherson’s (2012) *Test Bank*.

*Development of Polytomous Items and Their Q-Matrices.* The developed items are procedu-

ral in nature and must be adapted to the specific requirements of this study. In particular, the response formats and attribute specifications must be defined to match the proposed framework, in accordance with the criteria outlined in “Objective 1.” Based on this foundational information, Q-matrices associated with the test items will be constructed. To this end, the proposed TRACE-Q described in “Objective 2” will be implemented to generate provisional Q-matrices, which will then be evaluated by the research team and the two instructors, and further modified if necessary. The finalized Q-matrix will subsequently be used to analyze both the individual section-level datasets and the merged dataset, serving as a benchmark for assessing the performance of results obtained from the individual classes.

*Data Collection.* HUDM 5122 is offered every spring semester and is divided into three sections. Each section typically enrolls approximately 15–40 students. One section is delivered online, while the other two are taught in person. All sections are organized and delivered in a parallel manner, with shared instructional content and common learning goals. Student performance has traditionally been evaluated using homework assignments, one midterm exam, and a final project. Within this structure, the midterm plays a particularly important role in determining whether students have developed the foundational knowledge and skills necessary for successful completion of the final project. However, the midterm covers only the first half of the course material, which is generally less challenging than the second half. Consequently, the absence of a comparable assessment for the latter portion of the course has left some students insufficiently prepared to apply the associated knowledge and skills to the final project. In addition, because the primary goal of CD is to provide students with detailed information about their mastery and non-mastery of specific attributes or skills, the breadth of content covered by a traditional midterm is too wide to serve this purpose effectively. With these considerations in mind, we plan to replace the traditional midterm with three short quizzes (approximately 15 items each) during the Spring 2028 (and possibly 2029) semester, thereby maintaining a manageable number of targeted skills while providing more precise diagnostic feedback across the course.

The PI has discussed the project with the other two instructors and has obtained their consent and support to collaborate on this research. Further details regarding the data collection and the use of the data to evaluate the effectiveness of the proposed methods are provided below.

*Data Analysis.* First, for each section, examinees’ proficiency classes will be obtained using the proposed p-GNPC methods and, where available, their parametric counterparts. In a second step, responses from the three sections will be merged into a single dataset comprising about 105 students (based on enrollment in spring 2026). This combined dataset will then be re-analyzed with the same nonparametric and parametric methods applied before to the individual sections. Proficiency class estimates obtained from the merged dataset will serve as benchmark for evaluating the effectiveness and efficiency of the nonparametric and parametric methods at the classroom level. The rationale of this approach is that larger samples tend to produce more stable and reliable estimates, which are presumably closer to examinees’ true proficiency classes. Subsequently, profile-wise and attribute-wise agreement rates between proficiency class estimates obtained from the individual sections and those from the merged dataset will be computed separately for nonparametric and parametric methods. Comparing their agreement rates will allow for an assessment of the relative effectiveness of nonparametric versus parametric methods when analyzing small samples.

*Reliability and Validity of the Developed CD Tests.* Of course, these sophisticated techniques are meaningful only if the validity of the items and their q-vectors is well established. Procedures for assessing the basic psychometric properties of the tests, including reliability and validity, were

developed in the PI's CAREER grant proposal and are adopted here with necessary modifications. Test reliability is evaluated using an internal consistency index. Test validity is examined using Kane's Validity Argument Approach (Kane, 2001, 2004), which emphasizes the integration of multiple sources of validity evidence. This framework has been successfully applied to establish the validity of assessments developed by the Dynamic Learning Maps® Alternate Assessment System supported by the U.S. Department of Education (Clark & Karvonen, 2020).

Validation of the Q-matrices for CD tests is an integral component of the overall validity evaluation of the proposed methods. Given the small sample sizes typical of classroom settings, experts review will serve as the primary validation strategy. The research team will work closely with experts in math education and the instructors to examine the Q-matrices generated by TRACE-Q, discuss their substantive plausibility, and revise any potential misspecifications as needed.

## Supporting STEM Education and Research

### *Activity 1: Support Student Retention in STEM*

TC at Columbia University and UIUC have made sustained investments in research and practice to strengthen student success in STEM fields, with particular emphasis on retention. Both institutions prioritize evidence-based decision-making, resource allocation, and policies that improve academic trajectories, reduce attrition, and support timely degree completion. TC anchors this effort in rigorous scholarship through centers such as Community College Research Center and the Education for Persistence and Innovation Center (with which the PI is affiliated), which have pioneered national models for "Guided Pathways" and failure-based instruction, both of which target student persistence at critical transition points. UIUC supports STEM retention through coordinated initiatives that combine learning communities, mentoring, and early academic support in foundational STEM courses. Campus-wide efforts such as I-STEM promote evidence-based teaching practices and assessments to improve persistence and degree completion. Additional programs emphasize early research engagement and peer networks, particularly for first-generation STEM students.

We advance STEM retention through research engagement, coaching and mentoring, and targeted graduate training. Supported by her NSF CAREER grant, the PI previously recruited three PhD students at Rutgers University. Two of them have established successful careers and are currently employed at ETS, with multiple publications from their dissertations appearing in outlets such as *Psychometrika*, *British Journal of Mathematical and Statistical Psychology*, *Journal of Educational and Behavioral Statistics*, and *Applied Psychological Measurement*. Building on this record, the proposed work will further support persistence-to-degree and early-career continuity for emerging STEM researchers. More information is provided in the "Mentoring Plan."

### *Activity 2: Provide a Multidisciplinary Research Environment for Training Students*

This proposal is interdisciplinary, drawing on statistics, psychometrics, learning sciences, and information sciences, and will provide a strong foundation for PhD training in interdisciplinary research. Graduate students will engage in methodological development and applied work across these areas, contributing to multiple strands of the project. The research project is expected to support at least two doctoral dissertations. Students will meet regularly with the project team to discuss progress, key methodological decisions, and next steps. Faculty will provide guidance on academic development, including recommendations for relevant coursework to strengthen students' preparation. For instance, the Measurement, Evaluation and Statistics program at TC offers a rigorous MS in Applied Statistics that all doctoral students are encouraged to pursue; UIUC provides a strong campus-wide Graduate Certificate in AI. Additional interdisciplinary resources are available through institutional centers such as the Digital Futures Institute at TC, which supports research and training related to digital innovation. In sum, the project offers structured research and

training opportunities that prepare graduate students to conduct interdisciplinary work in modern educational measurement and learning analytics.

### Dissemination

#### *Activity 3: Publish Research Products and Host Professional Training Workshops*

The research outcomes of the proposed project will support four to six presentations at annual conferences and at least three peer-reviewed journal articles targeting audience across psychometrics, machine learning and AI, and educational data mining. As part of a broader dissemination strategy, we will also host training workshops at interdisciplinary academic conferences. Specifically, we plan to organize a one-day training workshop at major international venues such as AERA, NCME, IMPS, or AI in Measurement & Education (AIME) conference. The workshop will provide participants with a foundational understanding of the theoretical underpinnings of the proposed methods, along with hands-on experience applying these methods through the R software packages developed as part of the project.

#### *Activity 4: Build the Research Website and Host Free Webinars*

Nowadays, building an informative website is one of the most effective ways to communicate innovative research and its findings — a lesson learned by the PI through the development of the CATL-CD website, which documented her NSF CAREER research project. The website is currently in transition to TC. Planned future additions include technical reports and periodic updates on the progress of the proposed research, free downloads of newly developed R software, and a public forum to support discussion and the exchange of ideas. We would also reach out to the chairs of the NCME “Diagnostic Measurement” and “AIME” SIGIMIEs to explore opportunities for hosting free professional webinars.

### Broader Impacts

This project significantly lowers access barriers to the use of advanced CD assessments in classrooms by enabling diagnostic analysis of polytomous item responses underlain by dichotomous attributes or dichotomous responses underlain by polytomous attributes. At the same time, the proposed methods substantially reduce dependence on large samples and expert-intensive Q-matrix development. They support more timely, fine-grained feedback for instruction in small and resource-constrained educational settings where traditional CD approaches are infeasible. Project outputs will be disseminated through open-source extensions to the NPCDTools R package and the CATL-CD platform, with accompanying documentation and instructional materials for reaching out to non-specialists. By engineering a transparent and theory-driven integration of AI into assessment development, the project establishes a model for responsible usage of AI in education and broadens equitable access to diagnostic measurement tools for educators, researchers, and students.

### Research Timeline

Table 3 presents a schedule of the proposed objectives and activities over the 3-year period.

Table 3: Schedule of Proposed Research, Education, and Dissemination Activities

| Year | Research Plan 1 |        | Research Plan 2 |        | STEM   |        | Dissemination |        |
|------|-----------------|--------|-----------------|--------|--------|--------|---------------|--------|
|      | Obj. 1          | Obj. 2 | Obj. 3          | Obj. 4 | Act. 1 | Act. 2 | Act. 3        | Act. 4 |
| 1    | ✓               | ✓      | ✓               |        | ✓      | ✓      |               |        |
| 2    | ✓               | ✓      | ✓               | ✓      | ✓      | ✓      | ✓             | ✓      |
| 3    |                 |        |                 | ✓      | ✓      | ✓      | ✓             | ✓      |